Technical Document #793: Computer Vision - Comprehensive Overview

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Pose estimation detects human body keypoints in images. It is used in action recognition, animation, and sports analysis.

Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Object detection identifies and localizes objects in images. YOLO and Faster R-CNN are popular real-time object detection architectures.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Image classification assigns a label to an image from a fixed set of categories.
- Object detection identifies and localizes objects in images.
- Semantic segmentation classifies each pixel in an image into a category.